

An-Bao Xu (Ph.D.)

Associate Professor

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Education

- 2013.09-2017.07 **Ph.D.**, Hunan University, College of Mathematics.
- 2010.09-2013.07 **M.S.**, Guilin University of Electronic Technology, School of Mathematics and Computing Science.
- 2006.09-2010.07 **B.S.**, Xianning College (now Hubei University of Science and Technology).

Professional Information

Work Experience

- 2023.12.-present **Associate Professor**, Wenzhou University, College of Mathematics and Physics.
- 2017.07-2023.12 **Lecturer**, Wenzhou University, College of Mathematics and Physics.

Academic Visit

- 2016.11-2017.05 **Visiting Student**, Chinese Academy of Sciences.
- 2014.08-2016.08 **Joint PhD candidate**, Auburn University, USA.

Research Interests

- 1 Numerical Linear Algebra
- 2 Privacy Computing
- 3 Optimization

- 4 Deep Learning
- 5 Scientific Computing

Participation and Hosting of Foundation

2019.01–
2021.12 **Research on the Parameterized Thresholding Algorithm for Compressed Sensing and Matrix Completion**, *National Natural Science Foundation of China (Youth Program)* , (11801418).

[Host](#)

2014.08–
2016.08 **Joint PhD candidate**, *National Scholarship for Building High Level University*, (201406130046).

[Host](#)

Hosting of Research Topic (Special Issue)

Topic Editor, *Frontiers in Applied Mathematics and Statistics: “Advancements in Tensor Low-Rank Modeling for Data-Intensive Applications”*.

[Host](#)

Journal Articles (* Corresponding Author):

1. Hong Yang, **An-Bao Xu***, *Compressed randomized t-CSVD and its applications*, *Journal of Computational and Applied Mathematics*, 488(2026), 117766.
2. Jun Lei, Jiqian Zhao, Jingqi Wang, **An-Bao Xu***, *Tensor completion via leverage sampling and tensor QR decomposition for network latency estimation*, *Applied Intelligence*, 55:687(2025), 16p.
3. Frank Uhlig, **An-Bao Xu***, *Iterative optimal solutions of linear matrix equations for Hyperspectral and Multispectral image fusing*, *Calcolo*, 60:26(2023), 33 p.
4. Yongming Zheng, **An-Bao Xu***, *Tensor completion via tensor QR decomposition and $L_{2,1}$ -norm minimization*, *Signal Processing*, 189(2021), 10 p.
5. Dongxiu Xie, Hugo J. Woerdeman, **An-Bao Xu***, *Parametrized quasi-soft thresholding operator for compressed sensing and matrix completion*, *Computational and Applied Mathematics*, 149(2020), 24 p.
6. **An-Bao Xu***, Dongxiu Xie, *Low-rank approximation pursuit for matrix completion*, *Mechanical Systems and Signal Processing*, 95(2017), 77-89.

7. Dongxiu Xie, **An-Bao Xu***, Zhen-yun Peng, *Least-squares symmetric solution to the matrix equation $AXB=C$ with the norm inequality constraint*, International Journal of Computer Mathematics, 93(2016), 1564-1578.
8. **An-Bao Xu**, Zhen-yun Peng*, *Norm-constrained least-squares solutions to the matrix equation $AXB = C$* , Abstract and Applied Analysis, 2013 (2013), 10 p.

Other Papers and Preprints:

1. Jiahao Guo, **An-Bao Xu***, *High-performance privacy-preserving matrix completion for trajectory recovery*, arXiv preprint arXiv: arXiv:2405.05789 (2024).
2. Jiqian Zhao, **An-Bao Xu***, *Infrared small target detection via tensor $L_{2,1}$ -norm minimization and ASTTV regularization: a novel tensor recovery approach*, arXiv preprint arXiv:2402.18003 (2024).
3. Qianxi Wu, **An-Bao Xu***, *A biased deep tensor factorization network for tensor completion*, arXiv preprint arXiv:2105.09629 (2021).
4. **An-Bao Xu***, *Tensor completion via a low-rank approximation pursuit*, arXiv preprint arXiv:2004.08872 (2020).